On-Chip I_{DDQ} Testability Schemes for Detecting Multiple Faults in CMOS IC's

Changku Hwang, Mohammed Ismail, and Joanne E. DeGroat

Abstract—**In this paper we present two on-chip design-for-testability (DFT) schemes for CMOS IC's. One is for small circuits and the other for large circuits. Both schemes identify a faulty area on a chip with only a small area overhead for the additional circuitry and at most two extra pins. Moreover, if faults occur in different areas, multiple faults can also be detected with the proposed schemes. To demonstrate the ideas, DFT is incorporated in a 4-bit carry look ahead adder/subtractor (CLAAS) as well as a 16-bit arithmic logic unit (ALU). Simulation results are given.**

I. Introduction

It is well known that an analog fault which produces no logical error cannot be detected with a stuck-at test set. However, analog faults must be considered in chip testing since they degrade circuit performance and have the potential to become logical faults that generate a logical error [1], [2]. Furthermore, an analog fault leads to other abnormalities. One such abnormality is the current flow from power lines to GND in steady state response given a certain input test pattern. Detecting such current flow in the faulty location is the concept behind the quiescent power current testing (IDDQ) method [1], [3]–[8]. One of many current testing methodologies which detect such current is the distributed testing logic (DTL) [1]. It requires test patterns only for activation of faults [1], [9], but not for fault propagation [10] and requires no substantial modification in the circuit under test. Such modifications may put severe constraints on design styles and could prevent circuit optimization [11]. However, as pointed out in [7], the scheme in [1] requires a large number of extra transistors as each gate needs two extra transistors.

Another inherent limitation of VLSI testing is that it is impossible to observe certain nodes directly because of the complexity of modern VLSI circuits. So far very few on-chip VLSI testing schemes have been introduced which attempt to isolate the fully area. Most of the DFT schemes developed, e.g., [1], [7] only test whether or not the chip has a fault, but cannot locate the fault or even the faulty area.

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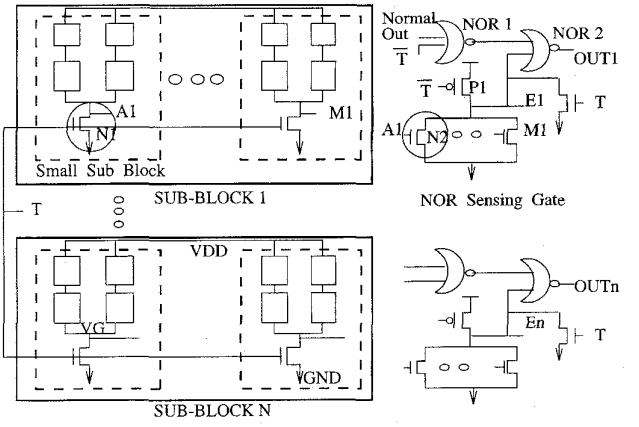


Fig. 1. The proposed Scheme 1.

This paper proposes two DFT schemes that can detect faulty areas with only a small amount of extra area overhead and at most two extra pins. The first scheme, suggested for detecting faults in a small circuit, is an improved version of the schemes reported in [1] and [7], while the second is a new scheme for detecting faults in a large circuit.

II. Proposed DFT Schemes

A. Scheme 1

Scheme 1 is illustrated schematically in Fig. 1. The entire design is divided into n SUBBLOCKS (SB) where n equals the number of outputs. Each SB is modified by inserting a n-channel MOSFET (N_1) for detecting the excessive current between GND and the n-channel network. In most cases, however, each SB should be divided into several small sub-blocks (SSB) each containing a number of logic gates. This reduces the resistive noise coupling [12] and the potential for changing the dynamic performance of the chip at the virtual ground (VG) during the normal chip operation. The partitioning of the device depends on its configuration.

The NOR sensing gate [designed with a p-channel pull-up MOSFET (P_1) and n-channel pull-down MOSFET's (N_2)] should be designed as static ratioed logic. The size of N_2 should be small because of the following reasons: small capacitive noise coupling [12], fast operation in test mode, and small area overhead. The output of the excessive current detecting circuitry is composed of two stage NOR gates. NOR1 and NOR2 are driven by the control

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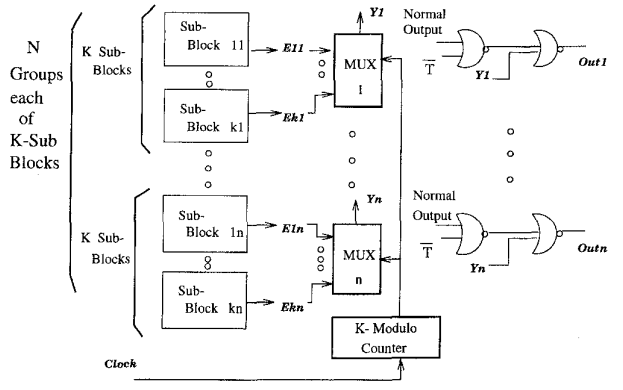


Fig. 2. The proposed Scheme 2.

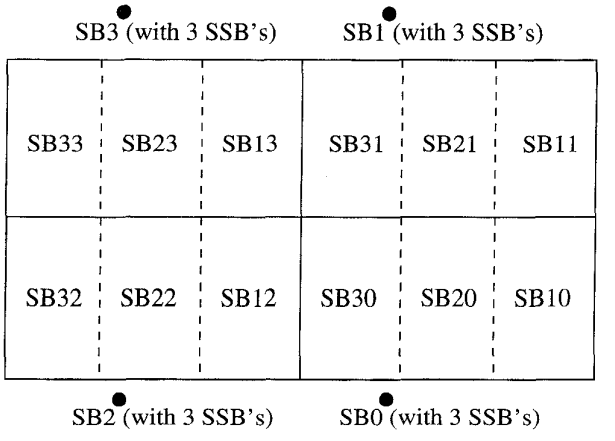


Fig. 3. Chip architecture for testing.

signal $\overline{T}$ and the output of the NOR sensing gate (error detecting signal E), respectively.

The operation of scheme 1 is as follows. In normal mode, T = logic 1, all N_1's are on while the P_1's of the NOR sensing gates are off. The N_1's allow current to pass through from the logic networks in the small-sub-blocks to GND and the E error sensing signals are maintained at logic 0. With T = logic 1, $\overline{T}$ = logic 0, and E_i = logic 0 the output, OUT_i, of the NOR2 is the normal output signal of SB_i.

In test mode, the control signal T is maintained at logic 1 while an appropriate input test vector is applied to the inputs to activate the possible faults. The control signal T is then switched to logic 0 and the outputs are observed. In a fault free case, there is no current flow from power supply to GND and the drain voltages of all N_1's are at logic 0. Accordingly, the E error detecting signals are logic 1 and the output of NOR2 represents logic 0. On the other hand, if there is a fault that provides a conducting path from the power supply to GND, the corresponding E error detecting signal is lowered to logic 0. The output of the NOR2 connected to this E error detecting signal becomes logic 1 indicating the presence of a fault in the corresponding *SB*.

There are three main advantages using Scheme 1:

1) Scheme 1 requires one extra control pin reserved for test/normal operation control (control signal T), while research reported in [1], [7], [9], and [10] requires two extra pins.
2) The extra area overhead needed for the fault detecting circuitry consumes less area than that reported in [1] and comparable to [7]. Moreover, if output stage buffers [13] are already used in the design, the number of extra transistors needed for the two stage NOR gates (NOR1 and NOR2) is decreased by using NOR1 and NOR2 for error detecting and as stage buffers.
3) The divided SB's are checked individually through their corresponding output and a faulty area is easily detected by observing the outputs. The effort and time to locate the fault is reduced as the fault is known to be located in the corresponding *SB*.

B. Scheme 2

Scheme 2 is a new approach for detecting the faulty areas using a counter and multiplexers. The design procedure of scheme 2 is shown in Fig. 2. For convenience, suppose that a k-modulo counter and the device having N outputs are used. First, the circuit is divided into $K \cdot N$ *SB*'s. The organization of *SB*'s and the excessive current detecting circuits are exactly the same as those of scheme 1. In Scheme 2, however, *NE* error detecting signal are the input into each of the K multiplexers. The k-modulo counter controlled by the signal, *Clock,* multiplexes the E error detecting signals to the NOR2's. Circuit operation is the same as that of Scheme 1. In the final step of test mode, however, each *SB* is tested individually, i.e., after the first clock, signals E_{11} to E_{1N} are selected and tested at the corresponding outputs. Any *SB*'s having faults output a logic 1 while the rest output logic 0. This testing continues up to the k_{th} clock, after which the signals E_{11} to E_{1N} are tested again. We note from the above results that Scheme 2 can detect faulty areas very precisely and only two extra pins (T & Clock) are required for testing. As a trade-off, Scheme 2 needs more transistors than Scheme 1.

III. Simulation

As a device under test (DUT), a 4-bit carry look ahead adder/subtractor was implemented using the *Magic Layout* editor and the *IRSIM* simulator. Its chip architecture for testing is shown in Fig. 3. A transistor stuck-on fault and a bridge fault were inserted intentionally in SB_{10} and SB_{30}, respectively. The transistor stuck-on fault was made such that it does not produce a logical fault so that it is not detected by a single stuck-at fault (SSF) test set. The

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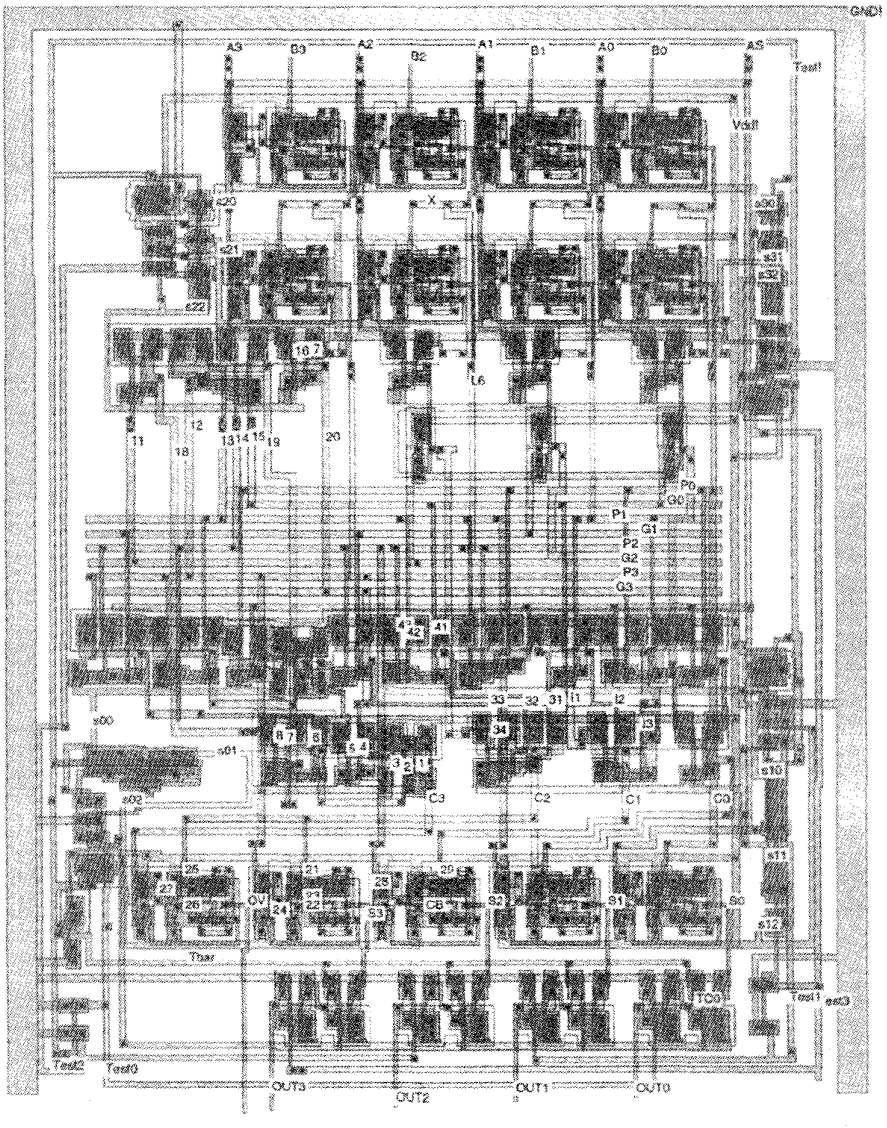


Fig. 4. The physical layout of the Carry Look Ahead Adder/Subtractor with Scheme 1.

TABLE I
TEST VECTORS AND OUTPUT RESPONSES

		tv_1	tv_2	tv_3
INPUTS	A	1010	1100	1111
	B	0001	0011	0011
OUTPUTS	Ideal case	01011	01111	10010
	Faulty case	01011	01111	00010

TABLE II
THE COMPARISON OF THE AREA INCREASE

		# of SB's	Area increase (%)
DUT1	Scheme 1	4	13.3
	Scheme 2	12	78.6
DUT2	Scheme 1	16	17.2
	Scheme 2	64	20.1

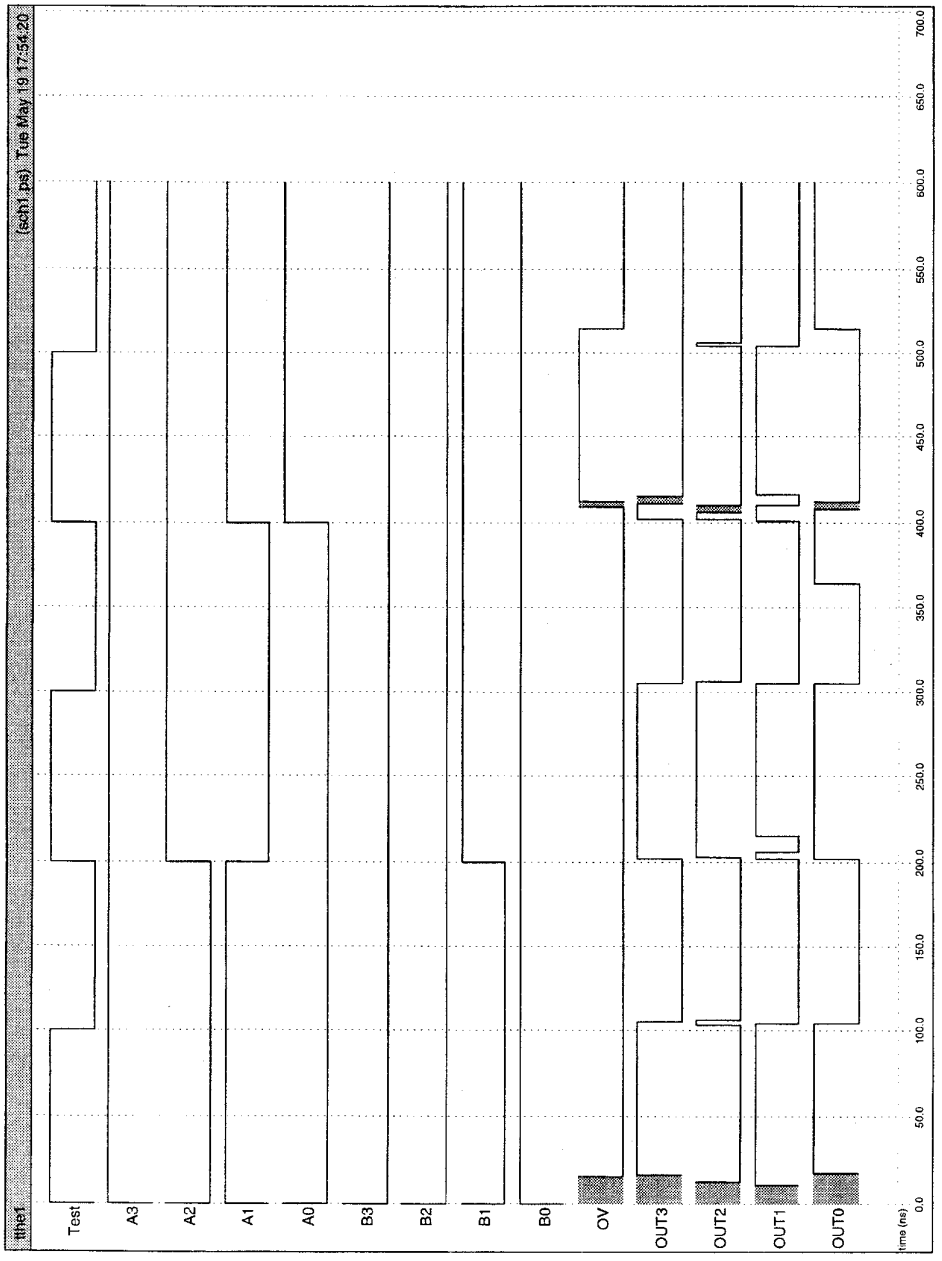
bridge fault produces a logical fault as well as current flow in the steady state response so that it is detected by SSF test set.

Three input test vectors used in the simulation and the outputs in steady state response are shown in Table I. Note that only tv_3 produces a logical error. As mentioned previously, a suitable input vector must be applied to activate a certain fault. Test vector tv_1 is an example of a test vector which neither produces a logical error nor faulty current flow. Test vector tv_2 only detects faulty current flow. Test vector tv_3 activates the bridge fault which causes both the logical error and faulty current flow.

A. Scheme 1

Since outputs are used for test, the chip is divided into four SB's where each SB has three SSB's. The physical layout and the timing diagram using tv_1, tv_2, and tv_3 are shown in Fig. 4 and Fig. 5(a), respectively. In the case of tv_1, since there is no conducting path from V_{DD} to GND,

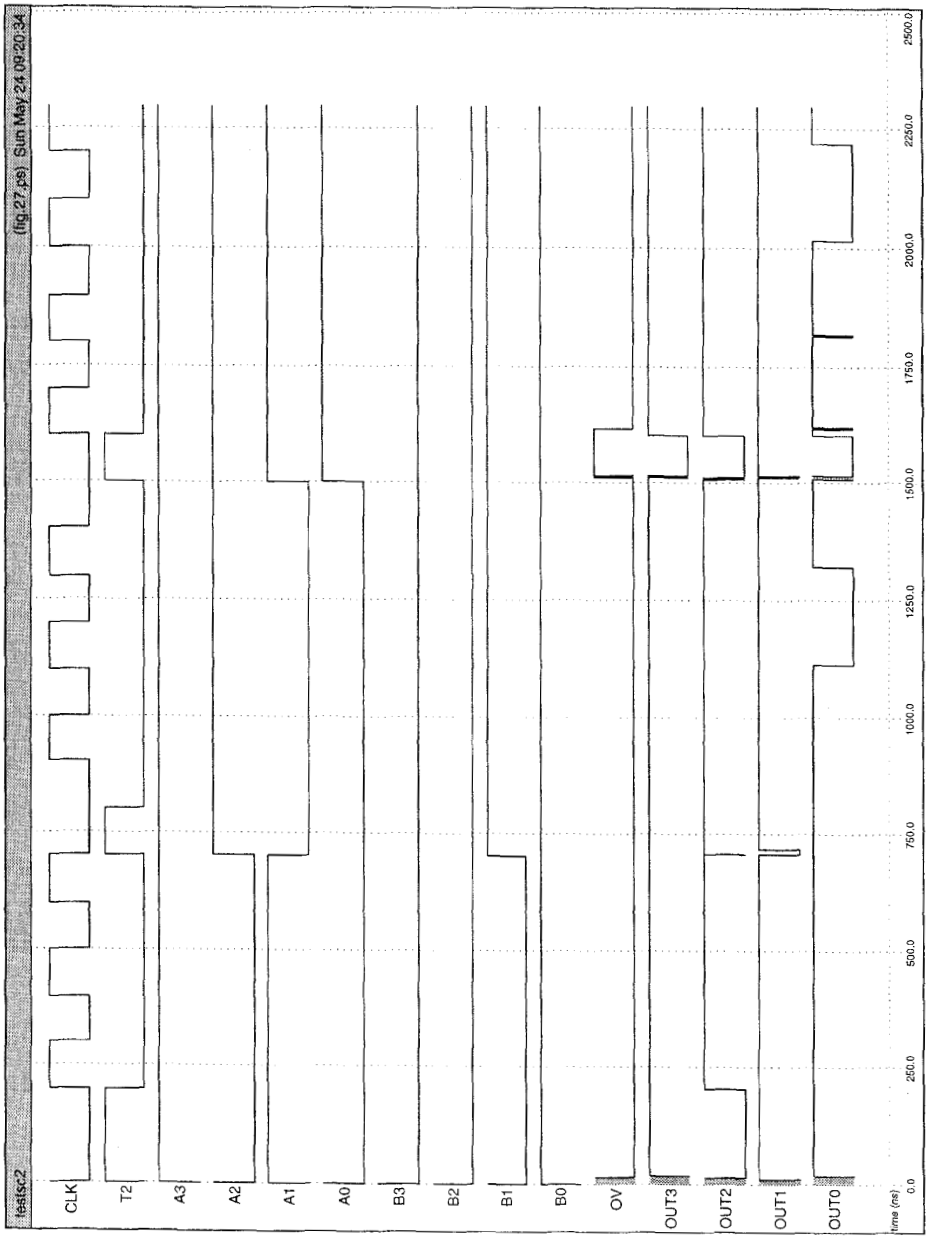
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(a)

Fig. 5. Timing diagrams of (a) Scheme 1.

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(b)

Fig. 5. (*Continued.*) Timing diagram of (b) Scheme 2.

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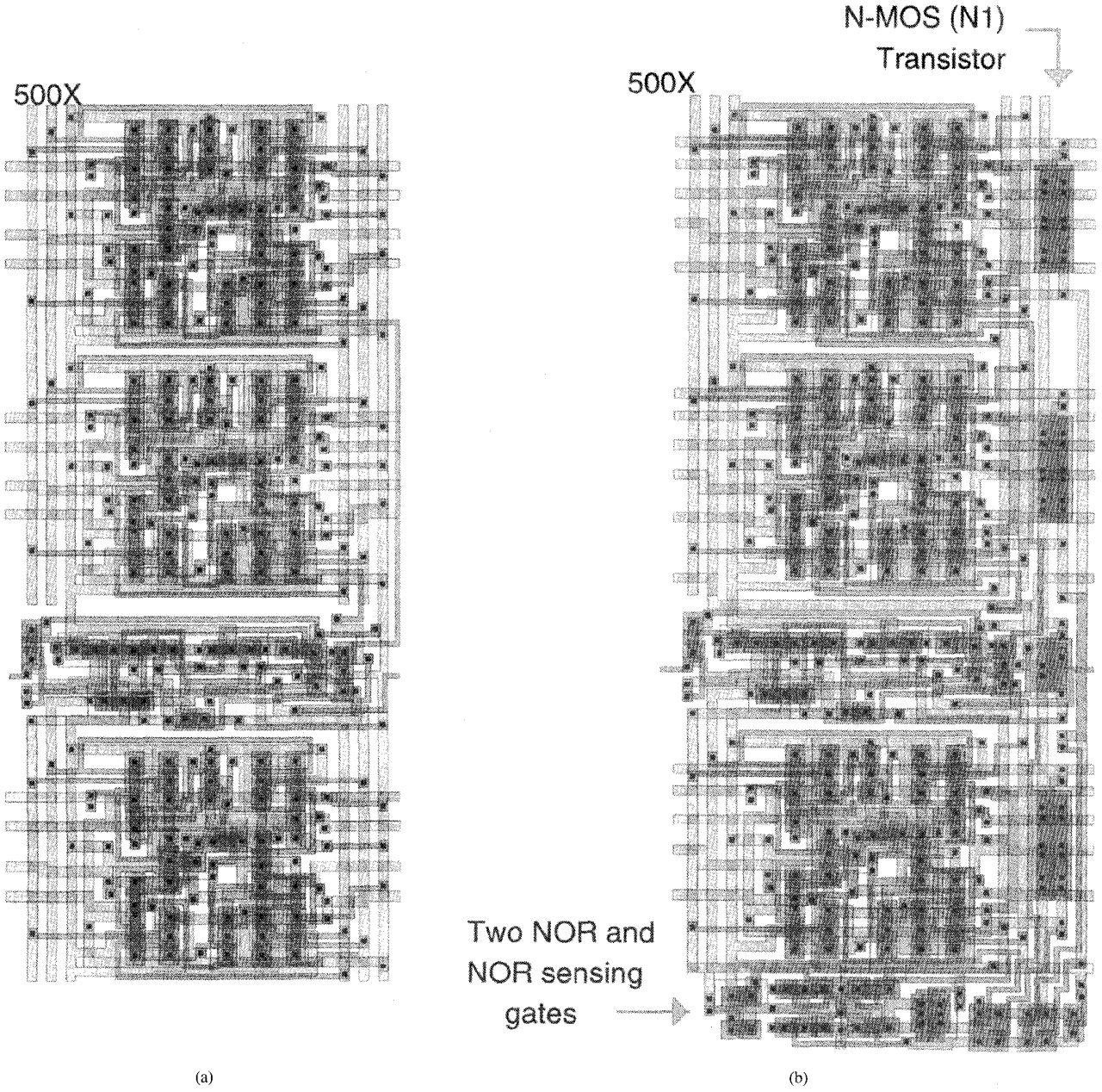


Fig. 6. Layout of a 1-bit ALU with (a) no DFT, (b) Scheme 1 and (c) Scheme 2. Corresponding 16-bit ALU architectures are built by cascading 16 identical 1-bit ALUs. The 1-bit block in (b) corresponds to one sub-block in the 16-bit architecture while the 1-bit block in (c) is partitioned into four sub-blocks; propagate generator, kill generator, carry chain, and result generator. The 2-bit counter in (d) is used for the 16-bit ALU with Scheme 2.

the outputs in the test mode are all at logic 0 (100 ns–200 ns). In the case of tv_2 and tv_3, output OUT_0 is at logic 1 (360 ns–400 ns and 520 ns–600 ns) as both tv_2 and tv_3 produce faulty current flow. The timing diagram indicates that SB_0 is a faulty area.

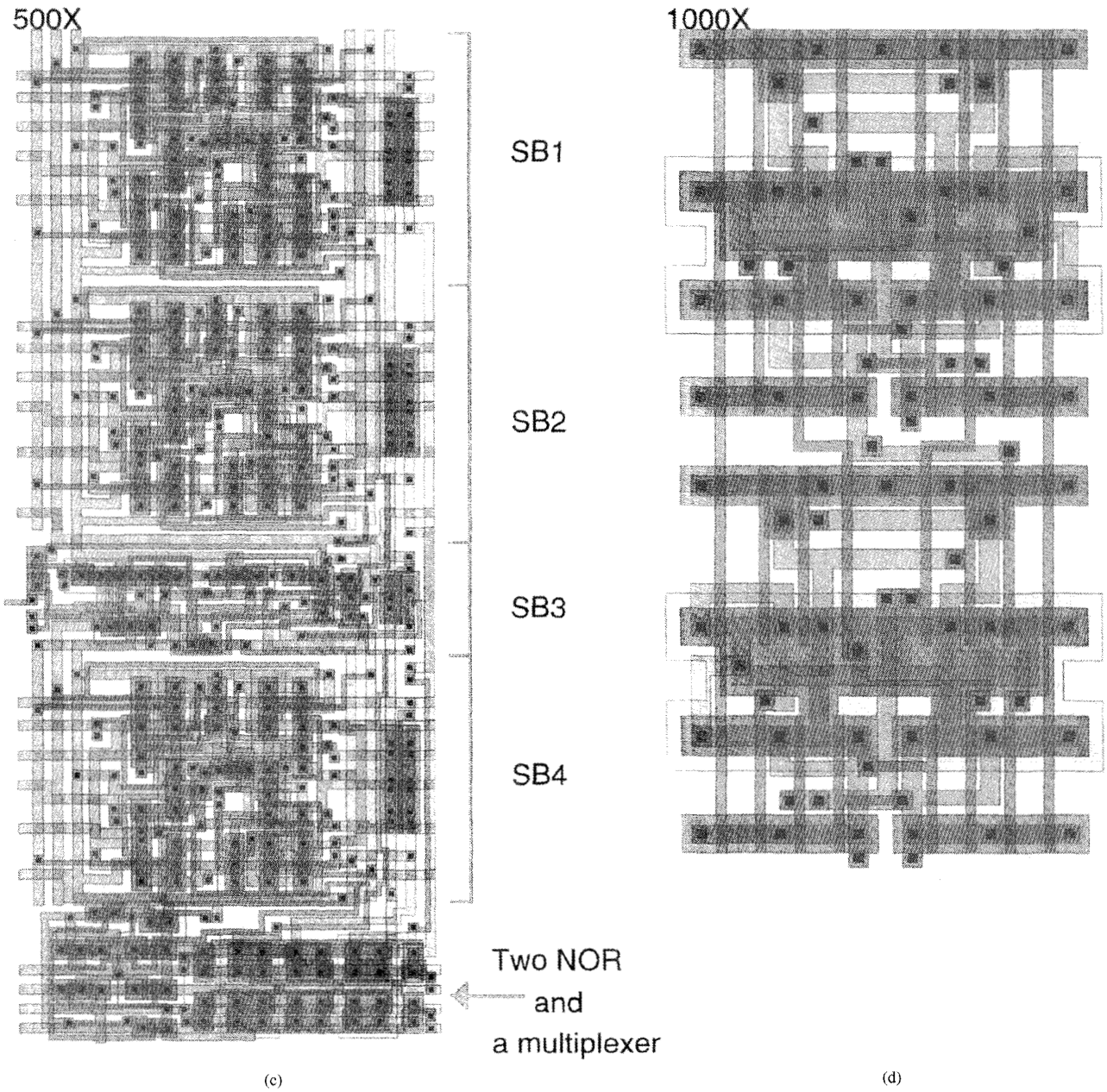
B. Scheme 2

Since DUT is divided into 12 SB's and four outputs are used, a standard Modulo-3 counter is used [14]. The timing diagram using the same test vectors as in the case of scheme 1 is shown in Fig. 5(b). Since NOR *Sensing gates* are not used in this simulation, the output value represents logic 0 if a fault is detected. In the timing diagram, one fault (Tr stuck-on fault) was found on OUT0 between 1114 ns and 1320 ns by test vector tv_2, while the other (Bridge fault) was detected between 2014 ns and 2220 ns by test vector tv_3.

Hence, one fault can be isolated to SB_{10} and another to SB_{30} by checking the number of clocks and the outputs indicating fault.

For the purpose of comparing the area overhead, both

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(c) (d)

Fig. 6. (*Continued.*)

Scheme 1 and Scheme 2 are applied to the 16-bit ALU (hereafter, DUT2) as well as the 4-bit carry look ahead adder/subtractor (hereafter, DUT1). The results are shown in Table II. Note that, after adding scheme 2, DUT1 grew by 78.6%. This is due to the fact that DUT1 is only at the SSI level (222 transistors) and of approximately the same complexity as the counter and multiplexer circuits used in the implementation of Scheme 2. On the other hand, when Scheme 2 was used with DUT2 (see Fig. 6 for details) which is at the MSI level (1728 transistors), the area increased only by 20.1%, even though the number of SB's increased from 12 to 64. This illustrates that as the circuit becomes larger, the area overhead to implement Scheme 2 will be less significant. It is very important to indicate that the number of SB's that can be tested using Scheme 2 increases exponentially according to the number of bits of the counter, while the growth rate of the area of the counter and multiplexers increases only linearly. Thus, Scheme 2 is best suited for circuits with greater complexity, MSI and greater.

IV. Conclusions

Two DFT schemes for detecting multiple faults in CMOS circuit are proposed in this paper. Especially meaningful is that the proposed schemes are capable of

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isolating the faulty areas in a circuit as well as detecting analog faults that cannot be revealed by conventional methods, i.e., no logical errors in steady-state response.

The main advantages of Scheme 1 compared to other existing DTL (distributed testing logic) designs are that it requires less area and less extra hardware as well as the ability for isolating faulty areas. However, its ability to isolate faulty areas is limited to the number of outputs of the circuit under test. Scheme 2 requires more area than Scheme 1. Using Scheme 2 the total number of SB's is not limited to the number of outputs but can be chosen appropriately for the size of the circuit. Scheme 2 is more suitable for testing LSI or VLSI circuits.

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